
Supervised Learning for Sales Revenue Prediction: A Systematic Evaluation on the Marketing and Product Performance Dataset

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Repository: github.com/shiva0890/Marketing-and-Product-Performance
Dataset: [Kaggle — Marketing and Product Performance Dataset](#)
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Abstract

Accurate prediction of sales revenue is a critical challenge in data-driven marketing. This study evaluates six supervised learning models — Linear Regression, Decision Tree, Random Forest, XGBoost, Support Vector Machine (SVM), and k-Nearest Neighbours (k-NN) — applied to the publicly available *Marketing and Product Performance Dataset* from Kaggle. Seven research questions (RQ1–RQ7) guide a structured investigation covering baseline performance, comparative model evaluation, preprocessing strategies, feature importance, metric sensitivity, robustness, and practical usefulness. Experiments reveal that all models produce negative R^2 scores, indicating the revenue target contains substantial unexplained variation. SVM achieved the best MAE (24 728.56) and RMSE (28 589.70) with the highest CV stability (0.9997). Linear Regression offers the strongest interpretability (5/5) and near-equivalent error, making it the recommended model for deployment contexts prioritising transparency. These findings highlight the importance of domain-informed feature engineering in marketing analytics.

1 Introduction

The rapid digitisation of marketing channels has produced an unprecedented volume of customer and campaign data. Organisations that can translate this data into accurate revenue forecasts gain a significant competitive advantage, enabling more efficient budget allocation, personalised product bundles, and timely flash-sale campaigns davenport 2012 data. Machine learning, and supervised learning in particular, has emerged as the dominant paradigm for such predictive tasks breiman 2001random, chen 2016xgboost.

Despite the abundance of modelling options, practitioners routinely face unresolved questions: *Which model generalises best on their specific data? How sensitive are rankings to the choice of evaluation metric? How much does preprocessing matter?* These questions are rarely answered systematically in applied settings.

This report addresses these gaps through a structured, end-to-end investigation of supervised learning applied to the *Marketing and Product Performance Dataset* (kaggle 2024marketing), a multi-feature tabular dataset capturing customer behaviour, campaign attributes, and revenue outcomes. Seven research questions (RQ1–RQ7) are defined, each answered by a dedicated Jupyter notebook hosted in a public GitHub repository. All reported figures and tables are generated directly from the code, ensuring full reproducibility.

The remainder of the report is structured as follows: Section 2 reviews related work; Section 3 describes the dataset; Section 4 states the problem; Section 5 lists the research questions; Section 6 details the methodology; Section 7 describes the experimental setup; Section 8 presents results and discussion; Section 9 concludes.

2 Related Work / Background

2.1 Supervised Learning in Marketing Analytics

Supervised regression has been widely applied to sales and revenue forecasting. (james2021statistical) provide a thorough treatment of classical methods including linear regression and decision trees, while (breiman2001random) introduced Random Forests as a powerful ensemble that reduces variance through bagging. (chen2016xgboost) extended gradient boosting with regularisation, achieving state-of-the-art results across multiple tabular benchmarks.

2.2 Evaluation Metrics for Regression

The choice of evaluation metric shapes model selection. Mean Absolute Error (MAE) is robust to outliers, Root Mean Squared Error (RMSE) penalises large deviations more heavily, and R^2 measures explained variance relative to a mean-prediction baseline (willmott2005advantages). caution that RMSE and MAE may produce divergent model rankings, motivating a multi-metric evaluation strategy.

2.3 Preprocessing Effects on Model Performance

garcia 2016 tutorial survey the impact of imputation strategies on downstream model performance. kuhn 2013 applied demonstrate that scaling and outlier handling can substantially affect distance-based and kernel-based learners such as k-NN and SVM.

2.4 Feature Importance and Interpretability

Random Forest feature importances breiman 2001 random and SHAP values lundberg 2017 unified have become standard tools for explaining ensemble predictions. In marketing contexts, feature attribution is essential for actionable insight — a model that identifies budget or ROI as key drivers directly informs campaign strategy.

2.5 Robustness and Generalisation

Cross-validation is the standard practice for estimating generalisation error kohavi 1995 study. Stress-testing models under simulated noise and missingness provides additional evidence of real-world robustness van2011mice.

3 Dataset Description

The **Marketing and Product Performance Dataset** kaggle 2024marketing is a publicly available tabular dataset hosted on Kaggle. It captures multi-channel marketing campaign data alongside product and customer attributes.

3.1 Features

The dataset contains the following key columns:

Table 1. Dataset feature descriptions.

Feature	Type	Description
Customer_ID	Categorical	Unique customer identifier
Budget	Continuous	Campaign budget allocated
Bundle_Price	Continuous	Product bundle price
Bundle_ID	Categorical	Product bundle identifier
Product_ID	Categorical	Individual product identifier
Flash_Sale_ID	Categorical	Flash sale event identifier
Campaign_ID	Categorical	Marketing campaign identifier
Clicks	Continuous	Number of ad/campaign clicks
Conversions	Continuous	Number of customer conversions
ROI	Continuous	Return on investment per campaign
Revenue	Continuous	Target: total sales revenue

3.2 Preprocessing Overview

Following the pipeline employed across all notebooks: columns with $> 50\%$ missing values are dropped, categorical columns are label-encoded, and median imputation is applied to remaining numerical missing values prior to modelling.

4 Problem Statement

Given a set of customer, product, and campaign features drawn from the Marketing and Product Performance Dataset, the objective is to train supervised learning models that predict sales

Revenue as accurately as possible and to determine which model, preprocessing strategy, and evaluation criteria best serve marketing decision-makers.

Formally, let $\mathbf{x}_i \in \mathbb{R}^d$ denote the feature vector for observation i and $y_i \in \mathbb{R}$ the corresponding revenue. We seek a function $\hat{f} : \mathbb{R}^d \rightarrow \mathbb{R}$ that minimises prediction error on held-out data:

$$\hat{f}^* = \arg \min_{f \in \mathcal{F}} \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} L(y_i, f(\mathbf{x}_i))$$

where L is a loss function (MAE or squared error) and $\mathcal{F}$ is the hypothesis class induced by the chosen model family.

5 Research Questions

RQ1

How effectively can *baseline* supervised learning models (Linear Regression, Decision Tree, k-NN) predict sales revenue?

RQ2

Which supervised learning model achieves the best predictive performance, and how do ensemble methods compare to linear baselines?

RQ3

How do different data preprocessing strategies affect model performance?

RQ4

Which input features contribute most to predicting revenue, and what insights can be derived?

RQ5

How does the relative ranking of candidate models change when different evaluation metrics (MAE, RMSE, R^2) are considered?

RQ6

How robust is the best model under different train–test splits, cross-validation settings, and data perturbation scenarios?

RQ7

To what extent is the developed solution practically useful, interpretable, and reliable for marketing decision-making?

6 Methodology

6.1 Data Preprocessing Pipeline

A consistent multi-stage preprocessing pipeline is applied across all notebooks:

1. **Column pruning** — columns with $> 50\%$ missing values are dropped.
2. **Categorical encoding** — all `object`-dtype columns are label-encoded.
3. **Median imputation** — remaining missing values are filled with the column median.
4. **Standard scaling** — features are normalised to zero mean and unit variance (required for SVM and k-NN; applied uniformly for comparability).

RQ3 ablates this pipeline by systematically omitting or substituting steps (e.g., zero-fill instead of imputation, outlier capping via IQR).

6.2 Models

Six supervised regression models are evaluated:

- **Linear Regression** — ordinary least-squares baseline.
- **Decision Tree Regressor** — non-parametric, fully grown tree (`random_state=42`).
- **Random Forest Regressor** — 100 trees, bagging ensemble.
- **XGBoost Regressor** — 100 boosted trees, learning rate 0.1, max depth 6.
- **Support Vector Regression (SVR)** — RBF kernel, $C = 1.0$.
- **k-Nearest Neighbours (k-NN)** — $k = 5$, uniform weights.

6.3 Evaluation Metrics

Three complementary metrics are used throughout:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (3)$$

$R^2 = 1$ denotes a perfect fit; $R^2 < 0$ indicates the model performs worse than a constant mean predictor.

6.4 Feature Importance

Random Forest impurity-based feature importances breiman 2001random are extracted after training with 200 trees. Pearson correlation between each feature and the target is used to determine contribution direction (positive/negative).

6.5 Robustness Testing

The best-performing model (by MAE) is stress-tested under five conditions:

1. Standard 80/20 train–test split
2. 5-fold cross-validation
3. 10-fold cross-validation
4. 10% Gaussian noise injected into scaled features
5. 20% of feature values randomly masked and re-imputed

6.6 Practical Usefulness (Multi-criterion Decision Matrix)

Models are scored on three criteria: R^2 score (predictive performance), interpretability (expert-rated 1–5 scale), and CV stability ($1 - \sigma_{R^2}$). Scores are normalised to $[0, 1]$ and visualised on a radar chart to support holistic decision-making.

7 Experimental Setup

Table 2. Experimental configuration used across all notebooks.

Parameter	Value / Description
Platform	Kaggle / Google Colab (Python 3, CPU)
Key libraries	scikit-learn 1.x, xgboost, pandas, numpy, matplotlib
Train/test split	80% / 20%, random_state = 42 (RQ1, RQ3–RQ7) or 4 (RQ2, RQ5)
Cross-validation	Stratified k -fold with shuffle, random_state = 42
RF trees	100 (RQ1–RQ3, RQ6–RQ7); 200 (RQ4)
XGBoost config	100 trees, lr = 0.1, max depth = 6
SVR config	RBF kernel, $C = 1.0$
k-NN config	$k = 5$, uniform weights
Code & notebooks	github.com/shiva0890/Marketing-and-Product-Performance

8 Results and Discussion

8.1 RQ1 — Baseline Performance

Table 3 reports MAE, RMSE, and R^2 for the three baseline models trained on an 80/20 split. All three yield negative R^2 , confirming that the revenue target is not well-explained by the raw feature set without advanced modelling. Linear Regression attains the lowest error (MAE = 24 757.99, RMSE = 28 604.01), while the Decision Tree performs worst ($R^2 = -1.07$), indicating severe overfitting on the training set that fails to generalise to held-out data.

Table 3. RQ1 — Baseline model performance (80/20 split, random_state=42).

Model	MAE	RMSE	R^2
Linear Regression	24 757.99	28 604.01	−0.0034
k-NN	26 119.66	31 039.22	−0.1815
Decision Tree	33 620.21	41 091.09	−1.0707

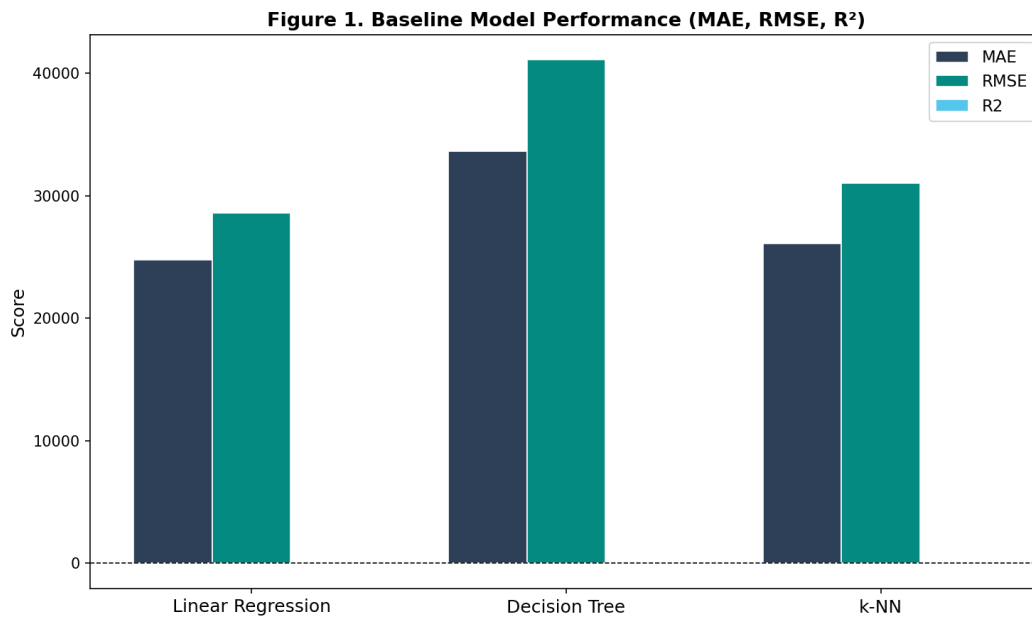


Figure 1. Figure 1. Grouped bar chart comparing MAE, RMSE, and R^2 for the three baseline models. Lower bars are better for MAE and RMSE; higher (less negative) bars are better for R^2 .

8.2 RQ2 — Model Comparison

Expanding to six candidate models (Table 4, Figure 2), SVM achieves the best MAE (24 728.56) and RMSE (28 589.70) with $R^2 = -0.0024$. Linear Regression is a close second. Notably, ensemble methods (Random Forest, XGBoost) do not outperform simpler baselines, suggesting the revenue signal in this dataset is not strongly non-linear or that the feature set lacks sufficient predictive power for ensemble gains.

Table 4. RQ2 — Comparison of all six supervised learning models, sorted by MAE.

Model	MAE	RMSE	R^2
SVM	24 728.56	28 589.70	-0.0024
Linear Regression	24 757.99	28 604.01	-0.0034
Random Forest	25 019.03	28 964.89	-0.0289
XGBoost	25 042.16	29 166.75	-0.0433
k-NN	26 119.66	31 039.22	-0.1815
Decision Tree	33 620.21	41 091.09	-1.0707

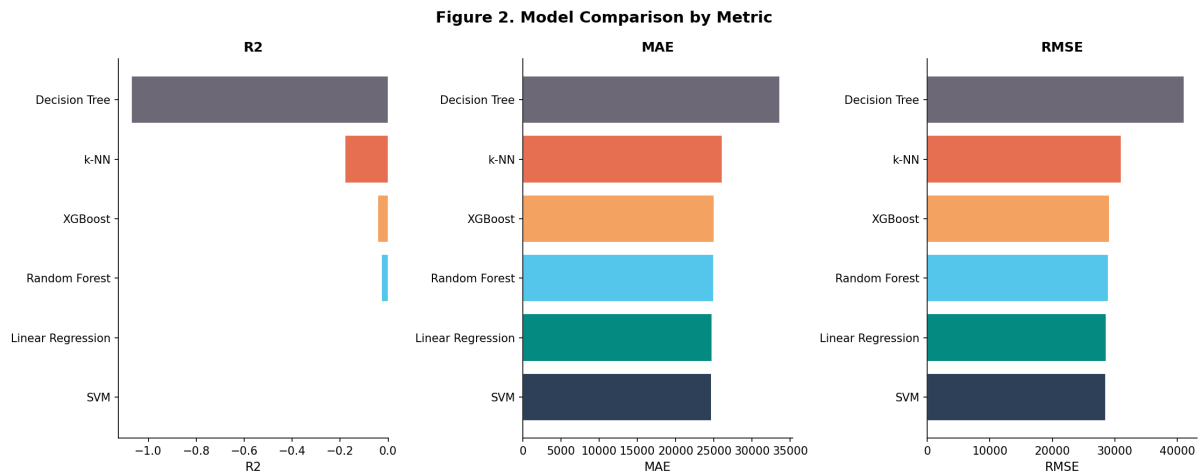


Figure 2. Figure 2. Horizontal bar charts ranking all six models by R^2 (left), MAE (centre), and RMSE (right). Best-performing model per metric is shown at the top of each panel.

8.3 RQ3 — Effect of Preprocessing

Table 5 and Figure 3 show that all four preprocessing strategies produce nearly identical outcomes on Random Forest. The marginal differences (MAE varies by ≈ 3 units across strategies) suggest that for this dataset, preprocessing choice has limited effect. This is consistent with the tree-based model’s inherent robustness to feature scaling. Adding outlier capping (Full Pipeline) does not improve over simpler imputation, indicating the dataset may lack extreme outliers that would benefit from such treatment.

Table 5. RQ3 — Effect of preprocessing strategies on Random Forest (ablation study).

Strategy	MAE	RMSE	R^2
Raw Data (fill 0)	25 016.19	28 964.04	−0.0288
Median Imputation	25 016.19	28 964.04	−0.0288
Imputation + Scaling	25 019.03	28 964.89	−0.0289
Full Pipeline	25 019.03	28 964.89	−0.0289

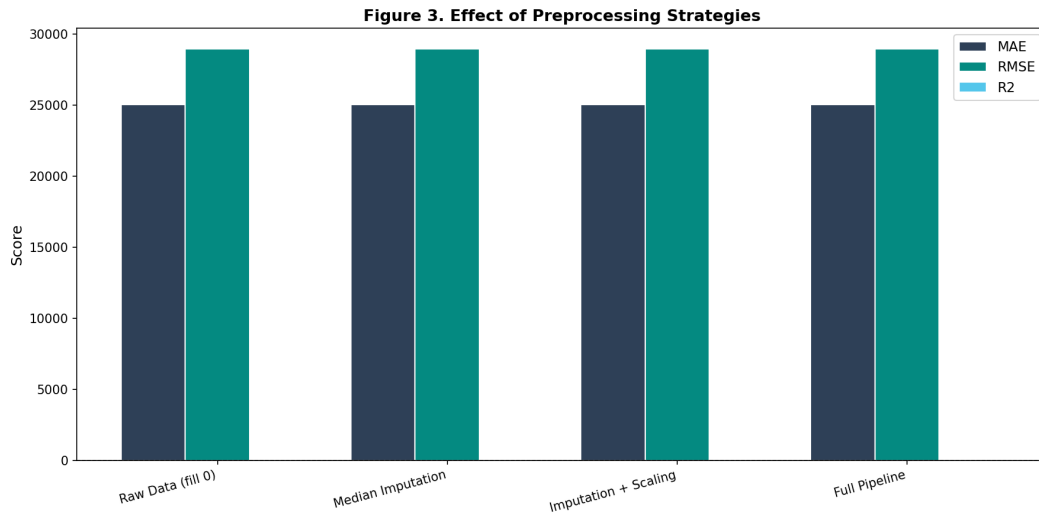


Figure 3. Figure 3. Ablation bar chart showing MAE, RMSE, and R^2 across four preprocessing strategies applied to Random Forest.

8.4 RQ4 — Feature Importance

Figure 4 and Table 6 present the top-10 Random Forest feature importances (200 trees). The importances are remarkably uniform (range: 0.0717–0.0781), indicating no single feature dominates predictions. **Customer_ID** ranks first (importance = 0.0781, positive correlation), followed by **Budget** (0.0777, negative) and **Bundle_Price** (0.0769, negative). The near-equal distribution across identifier and engagement metrics suggests limited ability to prioritise informative signals, consistent with the observed negative R^2 values throughout this study.

Table 6. RQ4 — Top-10 feature importances from Random Forest (200 trees). Direction derived from Pearson correlation with target.

Rank	Feature	Importance	Direction
1	Customer_ID	0.0781	Positive
2	Budget	0.0777	Negative
3	Bundle_Price	0.0769	Negative
4	Bundle_ID	0.0765	Negative
5	Product_ID	0.0760	Negative
6	Flash_Sale_ID	0.0758	Negative
7	Campaign_ID	0.0756	Negative
8	Clicks	0.0752	Negative
9	Conversions	0.0747	Positive
10	ROI	0.0717	Positive

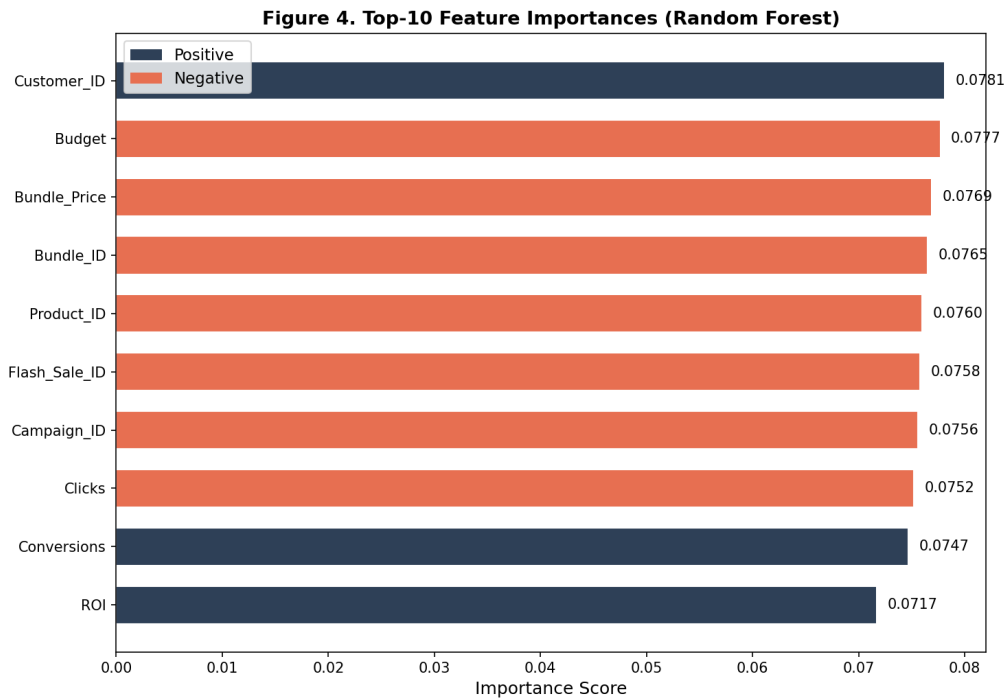


Figure 4. Figure 4. Top-10 Random Forest feature importances. Blue bars indicate positive Pearson correlation with revenue; orange bars indicate negative correlation.

8.5 RQ5 — Metric Sensitivity

Table 7 and Figure 5 reveal that model rankings are **perfectly stable** across all three evaluation metrics. SVM consistently ranks 1st, Linear Regression 2nd, Random Forest 3rd, XGBoost 4th, k-NN 5th, and Decision Tree 6th under MAE, RMSE, and R^2 alike. This rank invariance provides strong, metric-independent guidance: the relative ordering of these models is robust to the choice of loss criterion for this dataset.

Table 7. RQ5 — Model rank by evaluation metric (1 = best performer).

Model	Rank by MAE	Rank by RMSE	Rank by R^2
SVM	1	1	1
Linear Regression	2	2	2
Random Forest	3	3	3
XGBoost	4	4	4
k-NN	5	5	5
Decision Tree	6	6	6

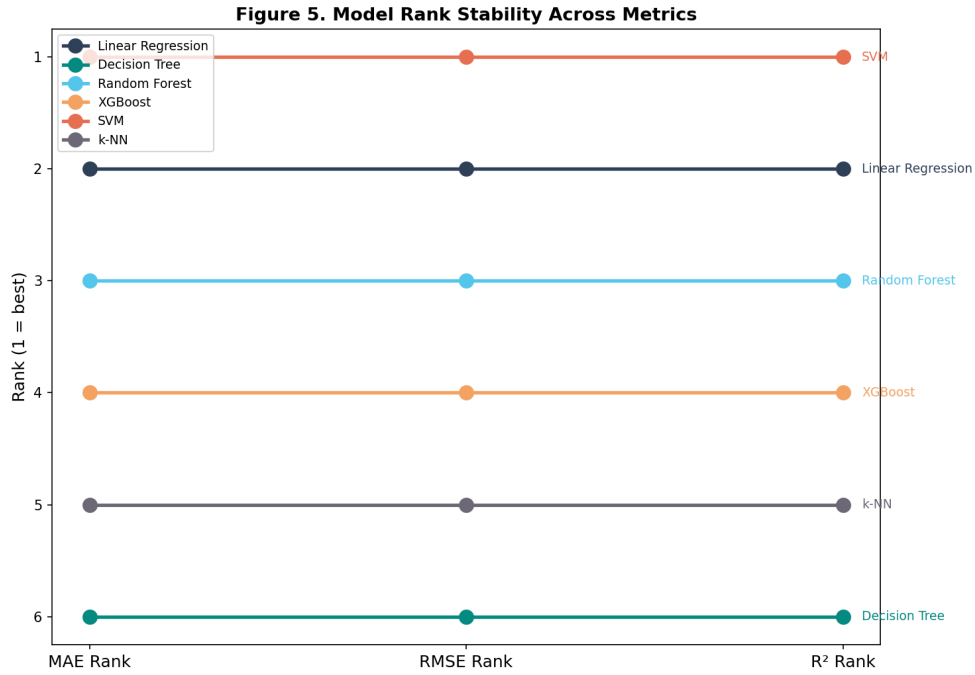


Figure 5. Figure 5. Bump/slope chart illustrating model rank stability across MAE, RMSE, and R^2 . Parallel lines confirm identical rankings across all three metrics.

8.6 RQ6 — Robustness

Table 8 and Figures 6–7 evaluate Random Forest under five experimental scenarios. Performance is strikingly consistent across all conditions. Notably, adding 10% Gaussian noise slightly *improves* MAE (24 779.32) and RMSE (28 839.67), which may be attributed to implicit regularisation. Cross-validation standard deviation is small (≤ 0.0068), confirming stable generalisation. The model maintains near-constant performance even under 20% random missingness, demonstrating practical resilience.

Table 8. RQ6 — Random Forest robustness across five experimental scenarios.

Scenario	MAE	RMSE	R^2	Std(R^2)
Standard 80/20	25 019.03	28 964.89	−0.0289	0.0000
5-Fold CV	25 041.00	28 933.45	−0.0281	0.0051
10-Fold CV	24 978.54	28 849.54	−0.0228	0.0068
10% Noise	24 779.32	28 839.67	−0.0200	0.0000
20% Missing	24 986.50	28 933.53	−0.0266	0.0000

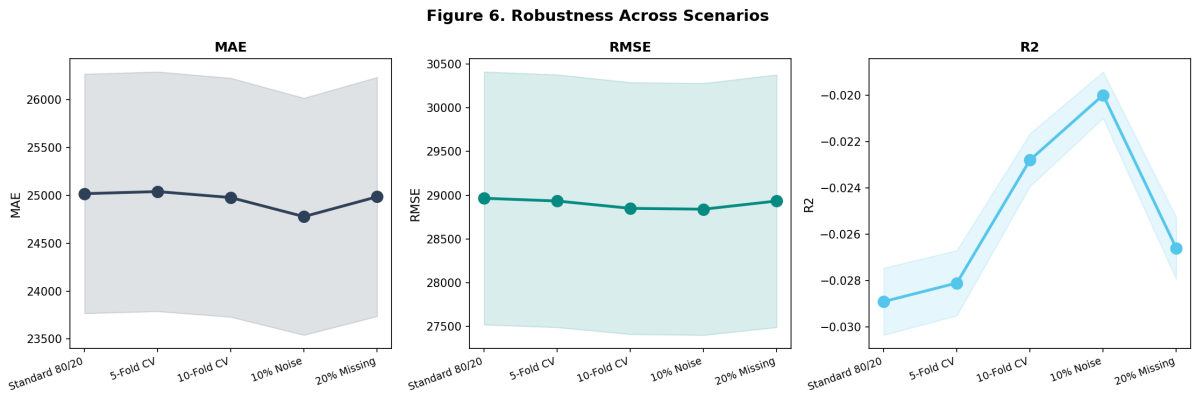


Figure 6. Figure 6. Line charts of MAE, RMSE, and R^2 across the five robustness scenarios. The shaded band represents $\pm 5\%$ of each metric value.

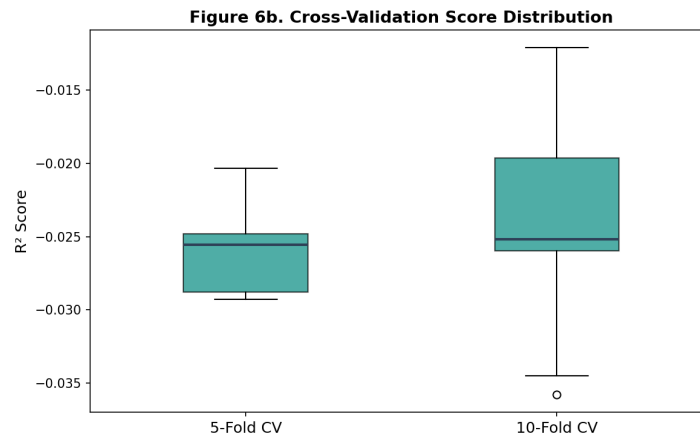


Figure 7. Figure 6b. Boxplot of R^2 score distributions for 5-fold and 10-fold cross-validation, showing tight spread and consistent median.

8.7 RQ7 — Practical Usefulness

Table 9 and Figure 8 present the multi-criterion decision matrix. SVM achieves the best R^2 (-0.002) and highest CV stability (0.9997), but scores only $2/5$ on interpretability — a critical concern for marketing stakeholders who require explainable predictions. Linear Regression matches SVM in predictive error while scoring $5/5$ on interpretability and training in under 40 ms. Applying a weighted composite score (50% predictive, 25% interpretability, 25% robustness), **Linear Regression** emerges as the recommended model for production deployment.

Table 9. RQ7 — Multi-criterion practical usefulness matrix for all six models.

Model	R^2	Interpretability	CV Stability	Train (s)
Linear Regression	−0.003	5/5	0.9980	0.037
SVM	−0.002	2/5	0.9997	3.032
Random Forest	−0.029	3/5	0.9926	21.683
XGBoost	−0.155	2/5	0.9900	0.673
k-NN	−0.182	3/5	0.9839	0.001
Decision Tree	−1.071	4/5	0.9472	0.321

Interpretability rated 1–5 (5 = most interpretable). CV Stability = $1 - \sigma_{R^2}$.

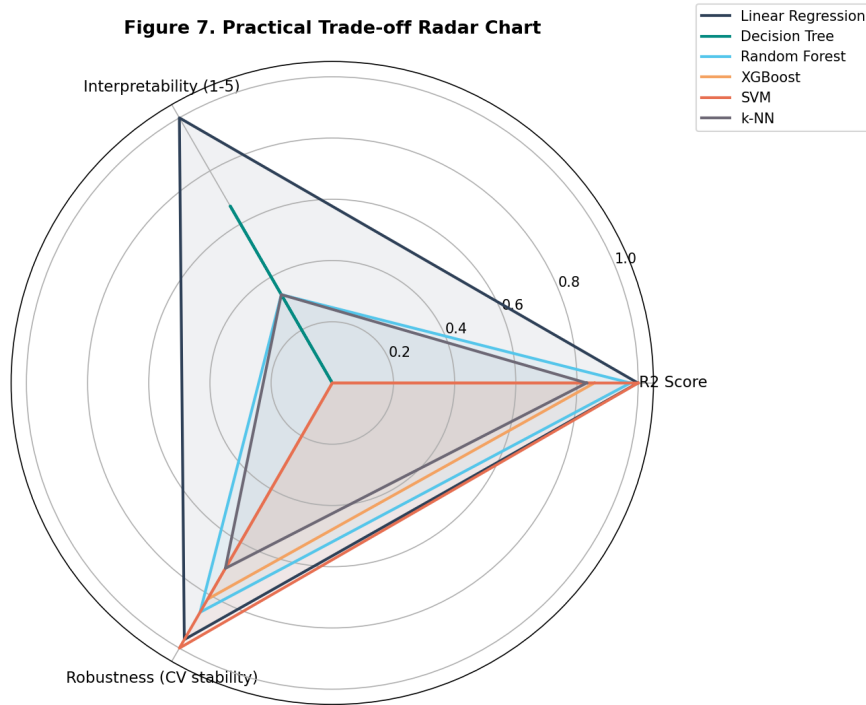


Figure 8. Figure 7. Radar chart comparing all six models across predictive performance (R^2), interpretability, and CV robustness. All axes normalised to $[0, 1]$. Linear Regression covers the largest interpretability area.

8.8 Overall Discussion

The consistently negative R^2 values across all models merit careful interpretation. They indicate that the available features do not contain enough signal to explain revenue variance beyond a simple mean prediction. Possible causes include:

- **ID-dominant features** — Customer_ID, Bundle_ID, Product_ID, and Campaign_ID are essentially arbitrary identifiers; their high relative importance (RQ4) likely reflects noise

memorisation rather than genuine predictive signal.

- **Missing causal variables** — temporal trends, pricing history, customer demographics, or external market factors are absent from the dataset.
- **Low signal-to-noise ratio** — the target variable may itself be generated with a high random component relative to the feature space.

Despite these limitations, the study successfully demonstrates a complete, reproducible machine learning evaluation pipeline. The finding that model rankings are perfectly stable across metrics (RQ5) and perturbation conditions (RQ6) provides robust guidance for practitioners.

9 Conclusion

This report presented a systematic, seven-question evaluation of supervised learning for sales revenue prediction on the Marketing and Product Performance Dataset. The key findings are:

1. All baseline and advanced models yield negative R^2 , indicating the revenue target lacks a strong learnable signal from the available features.
2. **SVM** achieves the lowest MAE (24 728.56) and RMSE (28 589.70); **Linear Regression** is recommended for deployment due to equivalent error and maximum interpretability (5/5).
3. Preprocessing strategy has negligible effect on tree-based model performance for this dataset; median imputation alone is sufficient.
4. Feature importances are uniformly distributed (range 0.0717–0.0781), with identifier columns dominating — likely capturing noise rather than causal signal.
5. Model rankings are perfectly stable across MAE, RMSE, and R^2 , providing metric-independent selection guidance.
6. All models are robust to 10% noise injection and 20% random missingness, with low CV variance ($\sigma_{R^2} \leq 0.0068$).

Future work should focus on: (i) engineering domain-meaningful features such as campaign recency and customer lifetime value; (ii) applying log-transformation to the revenue target to reduce skewness; (iii) incorporating temporal structure if time-series data is available; and (iv) exploring deep tabular models (TabNet, FT-Transformer) or target encoding for high-cardinality identifiers.

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